Groups determined by small incidence systems

Let $G = \mathrm{HS}$ in the acquired faithful action of degree 100. We retain its complete automorphism group N, of order 88 704 000, and the accepted list of coset geometries. Representatives below refer to that list. The source labels refer to the supplementary diagrams accompanying Leemans, *The residually weakly primitive geometries of HS*, Australasian Journal of Combinatorics 33 (2005), 231–236. Their identification with the acquired representatives is separately reviewed.

Theorem 1. *The full groups of representatives* $34, 48, 51, 61, 62$ *are as follows.*

Representative	*Source label*	$\mathrm{Aut}(\Gamma)$	$\mathrm{Cor}(\Gamma)$
34	*3.3*	G	$G \times C_2$
48	*4.34*	G	G
51	*4.27*	G	G
61	*4.24*	G	G
62	*4.23*	G	G

The assertion uses the acquired finite group model and the complete graph automorphism algorithm specified below. It is not a statement that the normalizer of a specified group action is automatically the full geometry group.

Lemma 2. *Let a flag-transitive coset geometry have a faithful specified group G, with parabolics P_i. Select all types whose cardinalities belong to a fixed set. Suppose every object is determined uniquely by the tuple of its shadows on the selected types. Suppose the induced incidence system on those types has full correlation group D, and the image of G in D is characteristic. Then every correlation of the original geometry normalizes its specified G action.*

Proof. The selected types form an intrinsic collection, so a correlation c restricts to a correlation of the induced system. For $g \in G$, characteristicity gives $h \in G$ whose restriction agrees with that of cgc^{-1}. The typed automorphism $h^{-1}cgc^{-1}$ fixes all selected objects. It fixes the shadow tuple of every remaining object and hence, by uniqueness, fixes that object. Thus $cgc^{-1} = h$. □

The argument is the same restriction argument used in the separately reviewed component for the geometries having a type of size 100. Here we select every type of size 176 or 1100. This selection, including repeated cardinalities, is checked against the actual subgroup indices. No type is selected by an arbitrary historical label.

Literal coset and shadow identifications. Write X for the union of the selected types. The certificate gives permutations on X for the ordered generators of the actual G. Their homomorphism is checked and is bijective onto its image E. For each selected type the specified base vertex has stabilizer exactly the image of P_i, and its orbit is the full type. Thus $P_i g \mapsto b_i^g$ is an actual coset bijection. For every ordered pair of selected types, the neighbours of b_i are checked to be precisely $b_j^{P_i}$. Every generator preserves every incidence and nonincidence row. Equivariance therefore proves the full incidence identification, not only agreement of local degrees.

The shadow of the base object of an arbitrary type j is the tuple $(b_i^{P_j})_i$. Its setwise tuple stabilizer in E is checked to be exactly the image of P_j, in all nineteen types of these five geometries. The orbit map then has fibres of size one. This proves the uniqueness required by the lemma, without constructing all objects of the larger types.

The complete upper bounds. The graph used for the induced incidence system has one marker for each selected type, joined to all objects of that type. Objects are coloured by their type cardinality. Markers have separate colours, also grouped by cardinality. A graph automorphism therefore permutes whole types of equal cardinality; conversely every correlation of the induced system extends uniquely to the markers. In particular, repeated types are not artificially distinguished by separate colours.

For representatives $34, 48, 51, 61, 62$ the graph sizes, including markers, are respectively $2379, 1278, 2379, 1278, 1278$. Complete nauty computations give orders $88\,704\,000$ in the first case and $44\,352\,000$ in the other four. All returned generators and all specified generators of G are checked on every coloured adjacency row. The group generated by the returned permutations has the reported order and has derived subgroup exactly E. Thus E is characteristic in each full graph group, and the lemma applies.

From normalizers to full groups. We now use the previously reviewed coset normalizer argument. Put $B = \bigcap_i P_i$; let S and T be the ordered and setwise family stabilizers in $N = \mathrm{Aut}(G)$. Once every full correlation normalizes G, flag transitivity and the base chamber give

$$|\operatorname{Aut}(\Gamma)| = |G : B||S|, \qquad |\operatorname{Cor}(\Gamma)| = |G : B||T|.$$

The calculation checks every type permutation. It discards a permutation only when subgroup orders, intersection orders, or a complete successive subgroup transporter proves it impossible. Every surviving transporter is checked on the entire family. This is exactly the already reviewed method, applied to five new literal families.

The resulting values are

Representative	$\lvert B\rvert$	$\lvert S\rvert$	$\lvert T\rvert$	$\lvert(T \cap G)/B\rvert$
34	720	720	1440	2
48	4	4	4	1
51	6	6	6	1
61	8	8	8	1
62	8	8	8	1

In every row $S \cap G = B$. Consequently the typed group is exactly the specified G. For row 34, the centralizer of G in the full correlation group is $(T \cap G)/B$, of order two. It is central and meets the typed group trivially. Their product has the full order, proving $\operatorname{Cor}(\Gamma) \cong G \times C_2$. The other four full groups already have order $|G|$ and hence equal G.

Verification and scope. The verifier checks actual stabilizers, base incidences and all graph generators; it does not repeat an ambient subgroup or geometry census.

The argument reuses elementary coset, shadow and normalizer ideas already established in the project. The new contribution here is their explicit application and the finite identifications. GAP's exact finite group algorithms and nauty's completeness are computational premises; no Lean verification, novelty, conjecture resolution or whole-manuscript completion is asserted. The other source obligations remain separate.